



A novel lightweight multi-attention fusion network for Alzheimer's disease detection

Jinhua Sheng^{a,b,*}, Ruilin Huang^{a,b}, Qiao Zhang^{c,d,e,**}, Tao Li^{a,b},
Miao Miao^{c,d,e}, Yan Lu^{a,b}

^a School of Computer Science, Hangzhou Dianzi University, Hangzhou, Zhejiang 310018, China

^b Key Laboratory of Intelligent Image Analysis for Sensory and Cognitive Health, Ministry of Industry and Information Technology of China, Hangzhou, Zhejiang 310018, China

^c Beijing Hospital, Beijing 100730, China

^d National Center of Gerontology, Beijing 100730, China

^e Institute of Geriatric Medicine, Chinese Academy of Medical Sciences, Beijing 100730, China

ARTICLE INFO

Keywords:

Alzheimer's disease
Structural MRI
Deep learning
Convolutional neural network

ABSTRACT

Alzheimer's disease (AD) poses a serious threat to the elderly and imposes heavy care burdens. Existing methods suffer from low data utilization and high computational cost. Here, we introduce a small-sample-oriented data utilization strategy on structural Magnetic Resonance Imaging (sMRI) and apply it with the lightweight Large-Small Network (LSNet) model for AD diagnosis. We propose the Multi-Granular Efficient Channel Attention (MECA) and Multi-Scale Aggregated Spatial Attention (MASA) modules, plus a Flexible Weight Fusion Attention (FWFA) mechanism to enrich attention features and overcome LSNet's single-pattern channel attention. The proposed modules were all compared with related works in ablation experiments and consistently outperformed the baseline. In particular, the LSNet modified with FWFA achieved slice-level accuracy, image-level decision accuracy, and Macro AUC of 78.14% (95% CI: 76.86, 79.42), 90.08% (95% CI: 88.86, 91.30), and 98.33% (95% CI: 98.06, 98.60), respectively, representing improvements of 3.03%, 5.88%, and 1.65% over the baseline. Furthermore, this study proposed a novel 2D MRI slice data processing pipeline and compared it with traditional methods in the field. Experimental results indicate that this slice synthesis strategy further enhances the model's learning capability. Overall, the improved model outperforms both the baseline and mainstream high-cost methods in terms of comprehensive performance and computational efficiency, demonstrating good generalizability and deployment flexibility, making it suitable for clinical devices with limited computing power or edge computing scenarios.

1. Introduction

Alzheimer's Disease (AD) is a neurodegenerative disorder characterized by subtle early symptoms, slow progression, and irreversibility. It is estimated that approximately 60%–70% of dementia cases eventually develop into AD [1]. Typical manifestations include memory decline, cognitive impairment, language and behavioral disturbances, severe mood fluctuations, and disruption of self-awareness [2]. As of 2024, about 6.9 million elderly individuals aged 65 and above in the United States suffer from Alzheimer's dementia. Without significant breakthroughs in prevention and treatment, the number of patients is

expected to rise to 13.8 million by 2060 [3]. As the disease progresses, AD patients become increasingly dependent on others, imposing considerable mental and physical stress on caregivers, as well as substantial social and economic burdens [4].

On the one hand, given the severe impact of AD, accurate and rigorous diagnostic assessments are necessary; on the other hand, the diagnosis of AD remains highly uncertain. Studies have shown that up to 23% of AD patients may be misdiagnosed [5,6]. The diagnostic process requires excluding other causes of cognitive decline, where clinicians typically employ advanced imaging techniques such as computed tomography (CT), magnetic resonance imaging (MRI), and positron

* Corresponding author at: Key Laboratory of Intelligent Image Analysis for Sensory and Cognitive Health, Ministry of Industry and Information Technology of China, Hangzhou, Zhejiang 310018, China.

** Corresponding author at: Beijing Hospital, 1 Dahua Rd., Beijing 100730, China.

E-mail addresses: j.sheng@ieee.org (J. Sheng), zhangqiao2941@bjhmoh.cn (Q. Zhang).

<https://doi.org/10.1016/j.bspc.2026.110183>

Received 30 June 2025; Received in revised form 2 March 2026; Accepted 28 March 2026

Available online 2 April 2026

1746-8094/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

emission tomography (PET) to identify AD-related features [7]. In recent years, rapid advances in computer science, especially machine learning and deep learning, have achieved breakthroughs in medical image analysis, brain region feature extraction, and disease prediction, holding promise for early AD screening. Intelligent algorithms can model high-dimensional features from multimodal imaging data such as structural MRI, PET, and fMRI, automatically capturing subtle lesions difficult for the human eye to discern, thus providing objective and precise clinical diagnostic support [8].

However, traditional machine learning-based AD diagnostic methods heavily rely on manual feature engineering, which often requires domain expertise that medical practitioners lack, leading to high entry barriers and limitations. Consequently, these methods are gradually being replaced by more intelligent deep learning approaches [9]. Although mainstream deep learning methods can automatically extract features end-to-end, their demand for large-scale annotated data conflicts with the scarcity of medical samples and privacy concerns. Furthermore, deep models generally involve substantial computational costs that ordinary hardware cannot afford, limiting their widespread clinical application [10].

To address these challenges, this study selects Large-Small Network (LSNet), a lightweight and efficient network, aiming to fully leverage limited medical imaging samples without significantly increasing computational burden. We first propose a data utilization strategy tailored for small-sample scenarios to maximize imaging information value. Then, from a radiomics perspective, transfer learning optimizations are applied to the LSNet architecture. Finally, we design two plus one innovative attention modules. Experiments demonstrate that our data usage strategy effectively mines sample information, and the three proposed modules significantly enhance LSNet's performance on AD classification tasks. Meanwhile, the overall approach maintains extremely low computational complexity, making it suitable for deployment on lightweight devices.

2. Related work

With the rapid development of information technology in recent years, machine-based decision-making has flourished in the medical field. Increasing numbers of methods have been proposed to assist medical practitioners in disease diagnosis and analysis. In the domain of Alzheimer's disease (AD) analysis assisted by unimodal structural MRI images, existing methods can be roughly divided into machine learning-based approaches and deep learning-based approaches.

2.1. Machine learning

In machine learning-based AD auxiliary diagnosis methods, researchers typically begin by manually extracting key features from brain images or clinical data using specific algorithms—such as texture, morphology, or gray-level distributions—to construct “feature engineering.” These high-dimensional features are then fed into classic classifiers like support vector machines (SVM), random forests, or logistic regression to automatically distinguish AD patients from healthy controls. For example, Xiao et al. [11] first extracted gray-level features from MRI and then used sparse logistic regression with generalized elastic net regularization for classification; Khan & Zubair [12] fused handcrafted/statistical multimodal features and applied ensemble learning methods based on multiple machine learning models for prediction; Basheera & Satya Sai Ram [13] employed the Leung–Malik filter bank to extract rotation-invariant texture features, built a visual dictionary, and classified with AdaBoost; Zhu et al. [14] used hippocampal functional connectivity strength as handcrafted features for hierarchical classification with SVM and other models; Lao & Zhang [15] generated low-dimensional features via NMF TDNet (essentially multilayer non-negative matrix factorization feature extraction), followed by SVM classification; Özçelik et al. [16] calculated pairwise correlations

between brain regions using the Pearson Correlation Coefficient (PCC) to generate functional connectivity matrices, then performed feature selection via the slime mold algorithm (SMA) and particle swarm optimization (PSO), and finally used KNN to classify individuals as AD or NC.

However, feature engineering occupies a central role in traditional machine learning, where these carefully designed and selected features themselves possess strong discriminative ability. The model's “intelligence” primarily manifests as pattern matching rather than autonomous reasoning. Therefore, to reduce manual intervention and enable models to automatically learn high-level representations from raw data, deep learning methods have gradually replaced traditional machine learning and become the current mainstream research paradigm [8].

2.2. Deep learning

Deep learning is an emerging field within machine learning aimed at enhancing machines' autonomous learning ability, enabling them to automatically learn and extract high-level abstract features from data. Deep learning-based AD diagnostic methods can be further categorized by input data format into 2D slice-based and 3D volume-based approaches.

Following anatomical conventions, 2D brain tissue slices obtained from MRI can be divided into three views: coronal (XoZ plane), axial (XoY plane), and sagittal (YoZ plane). Sun et al. [17] improved the ResNet architecture by using 2D MRI slices as input and optimizing residual modules to enhance diagnostic accuracy. Sina Fathi et al. [18] selected single-channel axial 2D MRI slices to build multiple convolutional neural network (CNN) models, combined with various ensemble schemes for comparison, and finally selected six best-performing classifiers to form an ensemble for fine-grained AD stage classification. Amir Ebrahimi-Ghahnavieh et al. [19] introduced recurrent neural networks (RNN) after CNN feature extraction to understand the relationship between slices in each subject's image sequence and made a comprehensive judgment based on all input slices. Their study extracted 48 axial slices, 72 coronal slices, and 57 sagittal slices per sample, which were further merged into 16 RGB axial images, 24 RGB coronal images, and 19 RGB sagittal images, respectively.

3D image-based methods typically input the entire MRI scan or segmented spatial voxels into 3D networks. Zeng, Li & Peng et al. [20] extended deep belief networks (DBN) into a multitask learning framework, inputting 3D volumetric features to simultaneously predict multiple clinical scores. Fan et al. [21] designed experiments based on a U-Net fully convolutional 3D network to jointly analyze segmentation and classification using 3D MRI. Katabathula, Wang & Xu et al. [22] proposed a lightweight 3D convolutional network focusing on global shape and texture features of the hippocampus, learned end-to-end. Yu et al. [23] incorporated higher-order pooling modules into a 3D GAN framework for tensorized generation and evaluation of MRI volumes. Guan et al. [24] designed a parallel attention-enhanced bilinear 3D network to directly learn discriminative features from whole-brain MRI volumes.

However, existing 2D MRI image-based methods either use only a single view, losing information from other views, or use three views simultaneously but without inter-view interaction. Moreover, multi-view approaches often assign a unified label for all views, making it difficult to highlight view-specific characteristics.

In addition, some studies first perform slice extraction and then randomly split the dataset, which results in slices from the same subject appearing simultaneously in both the training and testing sets. This leads to a certain degree of information leakage and affects the model's true performance.

Although 3D MRI-based methods can address the above issue, their large computational cost and difficulty in deployment are limiting factors. Therefore, drawing on previous work, we choose to conduct experiments using 2D slices, while improving data utilization and introducing innovations in network design. Fig. 1 illustrates the

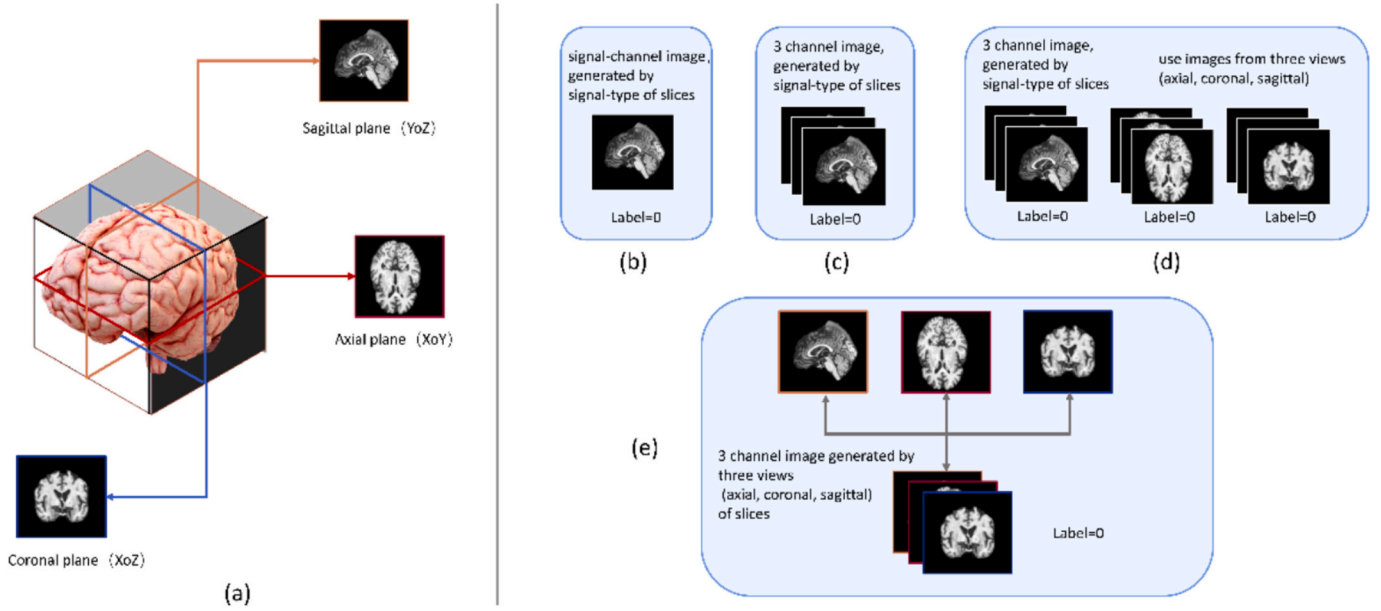


Fig. 1. (a) Principle of 2D MRI slice division; (b), (c), (d) 2D slices used in other studies; (e) 2D slices used in this study.

construction process of 2D MRI slices and provides a comparison between the traditional slicing strategy and the method proposed in this study, highlighting the differences in slice generation mechanisms.

In recent years, the attention mechanism has emerged as a key technique in deep learning, becoming an effective approach to enhance model learning capability. Its core idea is to adaptively assign weights, guiding the model to focus on feature regions or channels that are more discriminative for the current task, thereby suppressing redundant information and strengthening the representation of critical information. Introducing attention mechanisms in AD research can effectively direct the model to disease-related structures and enhance its ability to model pathological patterns. EROL Kina [25] proposed an EfficientNet + Squeeze Attention strategy to identify specific brain tissues in MRI images, achieving notable results; Abdelaziz et al. [26] employed multi-head self-attention and multi-head cross-attention for multi-modal scale fusion of MRI and PET, effectively highlighting discriminative brain regions.

Beyond incorporating attention modules to structurally modify traditional CNNs, Transformers—networks centered on self-attention capable of explicitly modeling global dependencies between features—have recently shown significant potential in Alzheimer's disease research. Hu et al. [27] proposed a VGG-TSwinformer model based on CNN and Transformer for short-term longitudinal studies of MCI. In the classification task of sMCI versus pMCI, the model achieved accuracy, sensitivity, specificity, and AUC of 77.2%, 79.97%, 71.59%, and 0.8153, respectively.

This study is built on a lightweight foundation, aiming to achieve superior diagnostic performance without excessively increasing model parameters or computational cost. To this end, instead of choosing the computationally heavy Transformer, we introduced a modular multi-attention fusion mechanism, which effectively enhances the network's representation of AD-related features and classification accuracy within an acceptable increase in computational overhead.

3. Materials and methods

The raw data and participant scans can be found in from the Alzheimer's Disease Neuroimaging Initiative (ADNI) (<https://adni.loni.usc.edu/>). Informed consent was obtained from the volunteer in accordance with the institutional review board policy. All methods were carried out in accordance with relevant guidelines and regulations. All experimental protocols were approved by the ethics committee at Beijing Hospital (2022BJYYEC-375-01) and the institutional review board (IRB) at

Hangzhou Dianzi University (IRB-2020001).

3.1. Large-small network

LSNet [28] is a lightweight, high-performance, and efficient network. Its design is inspired by the human visual system's "broad field of view" and "precise focus," which first captures wide-range contextual information through peripheral vision and then focuses on specific elements of interest via central vision to achieve detailed understanding.

3.1.1. LS (large-small) convolution

The LS convolution is the core component of LSNet. This convolution first employs a large-kernel convolution to capture extensive contextual information and model spatial relationships. Then, using the large-kernel output as parameters, it constructs a group-based small-kernel dynamic convolution to achieve focused attention on specific details within broad visual information. The LS convolution can be further divided into two parts: Large Kernel Perception (LKP) and Small Kernel Aggregation (SKA). Fig. 2 illustrates the computational principle of the LS convolution.

For an input feature x_i , let $N_P(x_i)$ denote the large-kernel perception region and $N_A(x_i)$ denote the small-kernel aggregation region, where the context perception range of $N_P(x_i)$ is larger than that of $N_A(x_i)$. The working process of LS convolution can be expressed as:

$$y_i = A(P(x_i, N_P(x_i)), N_A(x_i)) \quad (1)$$

where P denotes *perception*, referring to the process of extracting contextual features and modeling relationships among receptive field foci; A denotes *aggregation*, which integrates the perceived results from the previous step and fuses interactive information.

Large Kernel Convolution (LKP): For the input feature map $X \in R^{H \times W \times C}$, a pointwise convolution is first applied to project the image into a lower channel dimension. Then, for the dimension-reduced output x_i , a large convolution kernel with a receptive field of size $K_L \times K_L$ is used to extract wide-range contextual information centered at the focus. Finally, a pointwise convolution models spatial relationships to generate adjustable weights $w_i \in R^{H \times W \times C}$ for subsequent small-kernel focusing. The computation process of LKP can be mathematically expressed as:

$$w_i = P_{ls}(x_i, N_{K_L}(x_i)) = PW(DW_{K_L \times K_L}(PW(N_{K_L}(x_i)))) \quad (2)$$

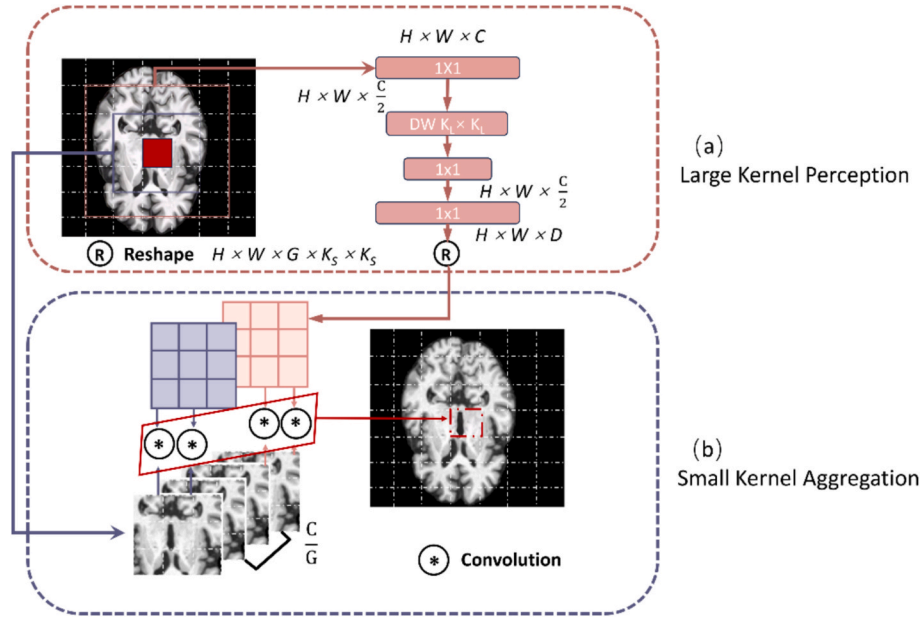


Fig. 2. Illustration of LS Convolution.

where PW denotes pointwise convolution, DW denotes depthwise convolution, and $w_i \in R^D$ represents the weights generated for x_i during the large-kernel perception process.

Small Kernel Aggregation (SKA): In this part, the original input feature map $X \in R^{H \times W \times C}$ is divided into G groups, each containing $\frac{C}{G}$ channels, with weight sharing within each group. For each x_i , the weights $w_i \in R^D$ learned by the LKP module are reshaped into group-wise small kernel convolution weights $w_i^* \in R^{G \times K_s \times K_s}$, where $K_s \times K_s$ is the size of the small convolution kernel. Then, the small kernels parameterized by w_i^* aggregate the contextual information within the neighborhood $N_{K_s}(x_i)$. Specifically, let x_{ic} denote the c -th channel of x_i ,

belonging to group g . Its aggregated feature y_{ic} is computed by:

$$y_{ic} = A_{is} \left(w_{ig}^* \cdot N_{K_s}(x_{ic}) \right) = w_{ig}^* \otimes N_{K_s}(x_{ic}) \quad (3)$$

where $\otimes$ represents the convolution operation.

3.1.2. LSNet model

LSNet is a network built around the core LS convolution operation, supplemented with a spatial attention module SE and a spatially encoded self-attention module (Multi-Head Self-Attention) to enhance model capability, and utilizes a feed-forward neural network (FFN) to perform

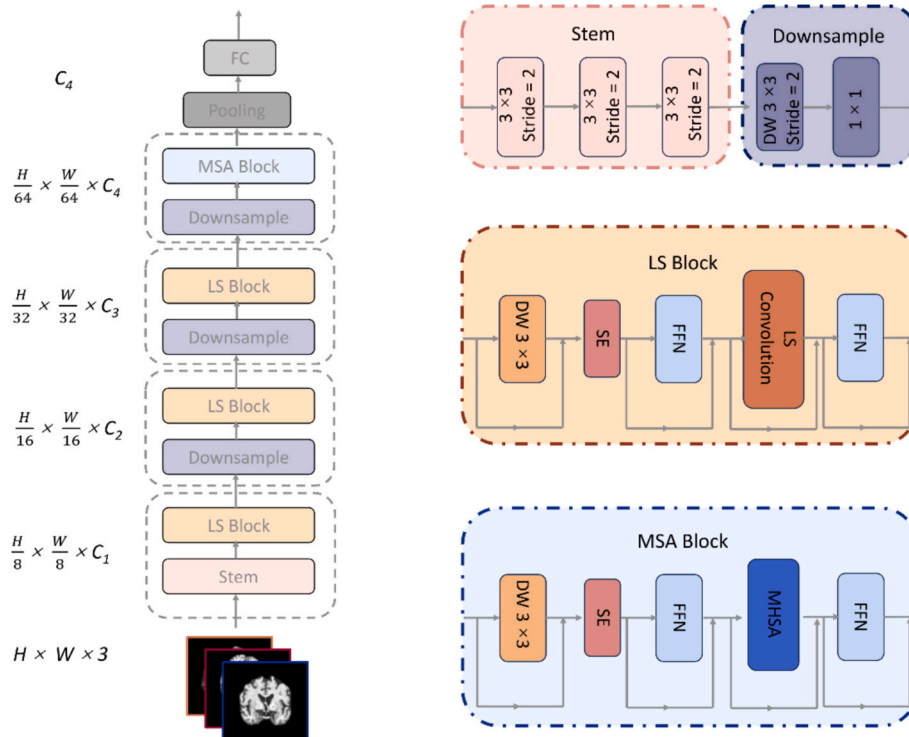


Fig. 3. Architecture diagram of the LSNet model.

channel mixing. Its architecture is illustrated in Fig. 3.

In our study, we observed that the attention module SE [28] used in LSNet is relatively basic. We believe that improving this module has the potential to further enhance the overall performance of LSNet.

3.2. Efficient Channel Attention (ECA)

Although existing channel attention methods improve performance, they inevitably increase model complexity. The Efficient Channel Attention (ECA) module was designed to overcome the trade-off between performance and complexity [29]. This module has multiple variants, all involving few parameters while delivering noticeable performance gains. The authors argue that dimensionality reduction can cause side effects on channel attention prediction, so avoiding dimensionality reduction is important for effective channel attention learning. Proper cross-channel interaction can maintain performance while significantly reducing model complexity. This study adopts two ECA variants, SE-Var2 and SE-Var3, which model channel weights using Hadamard product and MLP, respectively. In Fig. 4, we depict schematic diagrams of the two ECA modules and provide a comparison with the SE module.

3.3. Multi-Granular Efficient Channel Attention (MECA)

In channel attention modules such as SE and ECA, the compression step typically relies solely on global average pooling (GAP). This technique compresses the HH and WW dimensions into a single point, preserving only global statistics of each channel and failing to capture key information from local high-response regions.

For brain MRI images, different tissues (e.g., gray matter, white matter, cerebrospinal fluid) and lesion areas often exhibit significant local differences in intensity and texture [30]. These details are likely to be “smoothed out” by average pooling, reducing the attention mechanism’s sensitivity to critical regions like lesions or fine anatomical structures (e.g., sulci and gyri edges).

Therefore, this study adopts the key insight of the ECA module that dimensionality reduction may undermine channel attention, and integrates an innovative “multi-granularity feature fusion” design. Without applying dimensionality reduction for spatial attention

extraction, both Global Average Pooling (GAP) and Global Max Pooling (GMP) are used to construct diverse input features. Compared with the original ECA, this approach preserves the overall information within the receptive field while effectively capturing locally high-response features. For each input feature map, GAP and GMP are first applied independently, and their outputs are concatenated along the channel dimension. A linear layer is then employed to model implicit correlations across different granularities. In this way, the channel attention mechanism captures global brain image information while leveraging GMP to focus on anatomical boundaries, thereby enhancing sensitivity to local details such as lesion regions. Fig. 5 illustrates the architecture of FWFA, including the MECA and MASA modules.

Specifically, each input feature map $x_i \in R^{H \times W \times C}$ undergoes both GAP and GMP to produce $z_{iavg} \in R^{1 \times 1 \times C}$ and $z_{imax} \in R^{1 \times 1 \times C}$, respectively. These two pooled representations are concatenated along the channel dimension to form a more discriminative feature vector $z_i \in R^{1 \times 1 \times 2C}$:

$$z_{iavg} = \text{AvgPool}(x_i) \quad (4)$$

$$z_{imax} = \text{MaxPool}(x_i) \quad (5)$$

$$z_i = \text{Concat}(z_{iavg}, z_{imax}) \quad (6)$$

Next, a fully connected FC layer models inter-channel dependencies, replacing the traditional reduce-then-expand structure to reduce complexity while minimizing information loss. Finally, the output is passed through a sigmoid function to generate channel attention weights $w_i \in R^{1 \times 1 \times C}$, which are multiplied with the input feature map to obtain the refined output:

$$w_i = \sigma(\text{FC}(z_i)) \quad (7)$$

$$\hat{x}_i = x_i \odot w_i \quad (8)$$

3.4. Multi-Scale Aggregated Spatial Attention (MASA)

We observed that LSNet includes only a channel attention module but lacks spatial attention mechanisms. For medical images—especially brain MRIs—spatial structural information plays a crucial role in disease identification. Lesions often have strong spatial dependencies with surrounding tissues, which cannot be adequately captured by channel-

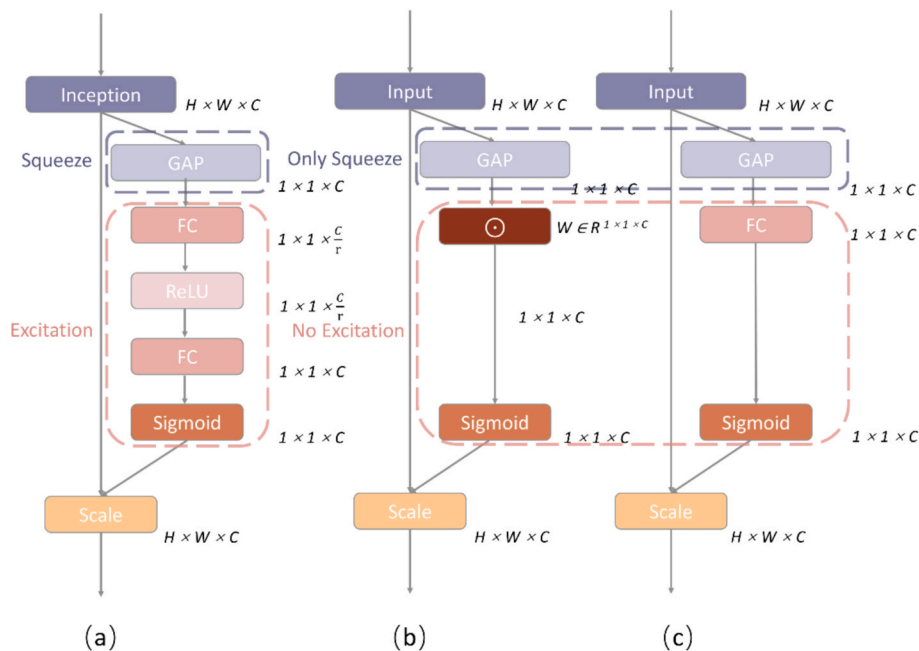


Fig. 4. Comparison of SE and ECA module designs; (a) SE module; (b) ECA-based SE-Var2 module; (c) ECA-based SE-Var3 module.

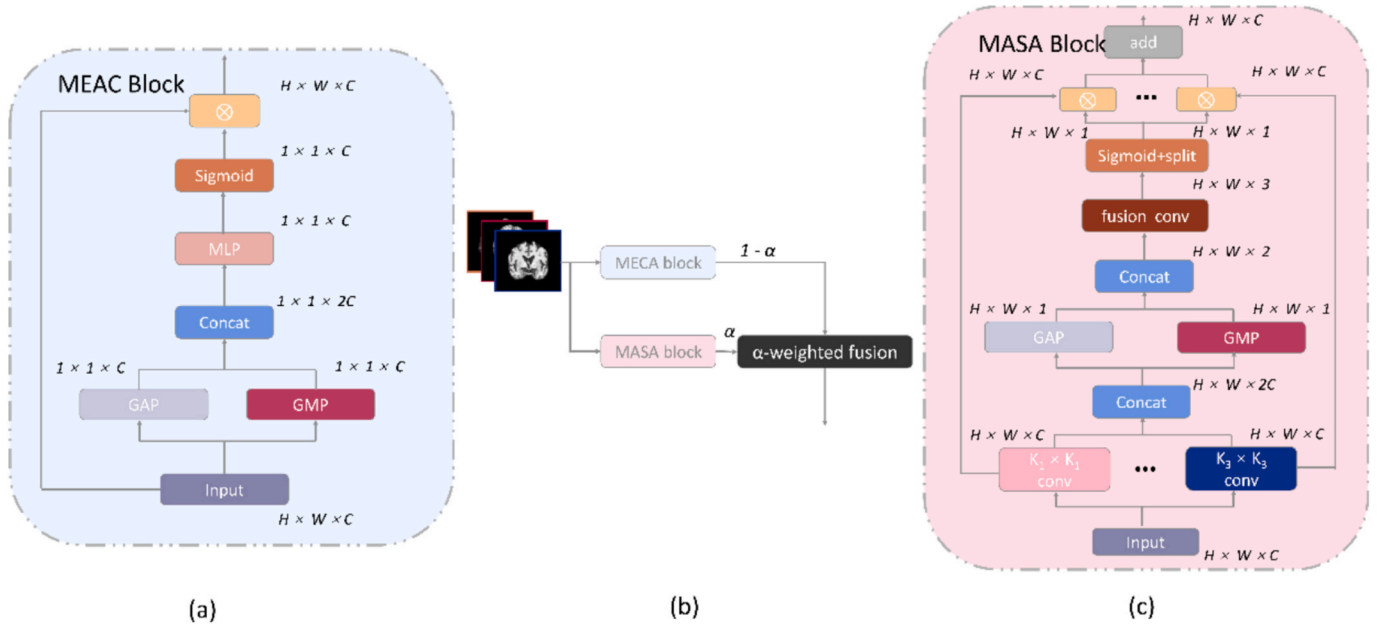


Fig. 5. Proposed modules in this study; (a) Structure of the MECA module; (b) Structure of the FWFA module; (c) Structure of the MASA module.

wise modeling alone. We hypothesize that incorporating spatial attention can enhance the model's sensitivity to key regions and improve overall performance.

Furthermore, the anatomical connections between brain regions are complex and not uniformly distributed. These inter-regional relationships exhibit regionality and hierarchy, with varying degrees of functional and structural coupling. Therefore, the context required to recognize lesions may differ depending on their location.

This issue can be addressed by introducing multi-scale spatial attention. In related studies, pyramid pooling [31] constructs multi-scale receptive fields through downsampling at different scales, while ASPP [32] replaces downsampling with dilated convolutions to make more effective use of spatial information. However, dilated convolution is relatively slow in computation, and both approaches rely primarily on multi-scale concatenation followed by convolution for feature fusion, resulting in a relatively single fusion strategy.

To overcome these limitations, we integrate the strengths of both methods by designing a multi-scale receptive field that combines standard convolution and dilated convolution, achieving a balance between representational capacity and computational efficiency. In addition, a spatial selection mechanism is introduced to adaptively learn the importance of information at different granularities. Specifically, multi-scale feature maps are concatenated along the channel dimension, followed by GAP and GMP for feature compression. The compressed features are then fused through convolution, and a Sigmoid function is applied to generate scale-specific spatial attention weights. Finally, a weighted summation is performed to obtain the final output.

For an input $x_i \in \mathbb{R}^{H \times W \times C}$, we first extract local visual information from the focus region using a standard convolution with kernel size $K_1 \times K_1$ (denoted as PA_{conv1}), while also introducing a dilated convolution with kernel size $K_2 \times K_2$ and $K_3 \times K_3$ and dilation rate r to capture long-range contextual dependencies (denoted as PA_{conv2} and PA_{conv3}). To reduce computational overhead, both convolutions are implemented as depthwise separable convolutions. The two features from different scales are then concatenated to form a multi-scale spatial feature $z_{ij} \in \mathbb{R}^{H \times W \times R}$:

$$PA_{conv1} = PW(DW_{K_1 \times K_1}) \quad (9)$$

$$PA_{conv2} = PW(DW_{K_2 \times K_2}^r) \quad (10)$$

$$PA_{conv3} = PW(DW_{K_3 \times K_3}^r) \quad (11)$$

$$z_{ij} = \text{Concat}(PA_{conv1}, PA_{conv2}, PA_{conv3}) \quad (12)$$

Here, DW denotes depthwise convolution, DW^r indicates dilated depthwise convolution with rate r , and PW denotes pointwise convolution.

Next, global average pooling (GAP) and global max pooling (GMP) are applied along the channel dimension of z_{ij} , reducing it from $H \times W \times R$ to $H \times W \times 1$ to retain both global semantics and salient spatial details. The outputs are concatenated to form a richer spatial representation $c_{ij} \in \mathbb{R}^{H \times W \times 2}$:

$$c_{ij} = \text{Concat}(\text{AvgPool}(z_{ij}), \text{MaxPool}(z_{ij})) \quad (13)$$

To facilitate cross-scale interaction, a convolutional layer FC_{fusion} is used to process c_{ij} , generating fused spatial features. A sigmoid activation σ is applied to obtain a total weight map $w \in \mathbb{R}^{H \times W \times 3}$. This is split into two spatial weights $w_j \in \mathbb{R}^{H \times W \times 1}$ for each scale:

$$w = \sigma(FC_{fusion}(c_{ij})) \quad (14)$$

$$w_j = \text{Split}(w), j = 1, 2, 3 \quad (15)$$

These weights are applied to the corresponding feature maps, and a weighted sum is computed to obtain the final fused spatial attention feature $\hat{x}_i \in \mathbb{R}^{H \times W \times C}$:

$$\hat{x}_i = PA_{conv1} \odot w_1 + PA_{conv2} \odot w_2 + PA_{conv3} \odot w_3 \quad (16)$$

3.5. Flexible Weight Fusion Attention (FWFA)

Channel attention modules typically use global pooling to compress spatial dimensions into a single point, modeling dependencies between channels. However, this "compression to a point" strategy is too rigid for medical images, especially complex anatomical structures like brain MRIs, where critical details such as singularities, folds, and boundaries may be lost.

To this end, we introduce a spatial attention mechanism as an effective complement to channel attention, forming a multi-attention architecture. In this domain, the classical Convolutional Block

Attention Module (CBAM) [33] adopts a sequential structure: channel attention is first computed from the input features, followed by spatial attention modeled based on the refined features. In contrast, we construct channel attention and spatial attention branches in parallel, modeling the input features independently to ensure the effectiveness of both attention mechanisms. A learnable weight parameter α is introduced between the two branches, enabling the model to adaptively adjust the relative importance of channel and spatial attention during training, thereby achieving more flexible feature fusion. Furthermore, multi-scale receptive field modeling is incorporated into the spatial attention module (MASA), which is an advantage not present in the standard CBAM. Fig. 5 illustrates the overall architecture of FWFA, as well as the detailed structural diagrams of MECA and MASA, which serve as its input components.

Parameter α essentially serves as a parameterized gating mechanism and, similar to convolutional kernel parameters, is dynamically updated through gradients. The initial value of α is set to 0.5, corresponding to an initial weight ratio of 1:1 between channel attention and spatial attention. Its value is constrained within the (0,1) interval to ensure that the sum of the two attention weights always equals 1, thereby maintaining the stability and interpretability of the fusion process.

The final attention output is:

$$out_{se} = MECA(x_i) \quad (17)$$

$$out_{pa} = MASA(x_i) \quad (18)$$

$$out_{fusion} = \alpha out_{pa} + (1 - \alpha) out_{se} \quad (19)$$

Here, MECA and MASA are the improved channel and spatial attention modules proposed in this paper.

4. Experiments

4.1. Data

The dataset used in this study comes from the Alzheimer's Disease Neuroimaging Initiative (ADNI), a multicenter longitudinal cohort study launched in 2004 and jointly funded by the NIH, pharmaceutical companies, and academic institutions. This open-access database includes multimodal medical data, providing a rich source of imaging and clinical information for early diagnosis and disease progression tracking. It has also supported the application of various machine learning and deep learning methods in neurodegenerative disease detection. For more information, visit: <https://adni.loni.usc.edu/>.

4.1.1. Data selection and preprocessing

This study uses structural MRI (sMRI), a medical imaging technique for obtaining high-resolution anatomical brain scans. It is particularly suited for analyzing brain changes associated with Alzheimer's disease by visualizing the morphology and volume of gray matter, white matter, and ventricles. We used 1.5 T T1-weighted structural MRIs for three subject groups: CN (cognitively normal), MCI (mild cognitive impairment), and AD.

To ensure better differentiation between CN and MCI groups, only CN samples with Mini-Mental State Examination (MMSE) scores ≥ 27 were selected. Also, to address sample imbalance—a common issue in ADNI-based studies—we applied class-balancing strategies to maintain a 1:1:1 ratio among CN, MCI, and AD in both training and testing sets.

Due to scanner differences, subject variability, and environmental conditions, raw MRI scans may suffer from distortions, brightness inhomogeneities, and noise. Furthermore, non-brain tissues such as the skull and eyes must be removed, and anatomical differences across individuals make voxel-based alignment necessary. The following preprocessing steps were applied:

Gradwarp correction – Corrects image distortions caused by gradient nonlinearities.

B1 inhomogeneity correction – Adjusts for signal intensity variations.

N3 bias field correction – Further reduces low-frequency intensity nonuniformities after Gradwarp and B1 correction.

Skull stripping – Removes non-brain tissues from the scan.

Registration to MNI space – Aligns all brain images to a common anatomical template.

Steps 1–3 were performed by the official ADNI preprocessing pipeline, while steps 4 and 5 were implemented using FreeSurfer in this study. A few corrupted scans were excluded during preprocessing. The final dataset contains 1193 samples, including 401 CN, 402 MCI, and 390 CE subjects from the ADNI1 phase. The statistical information of the experimental data is shown in Table 1.

4.1.2. Slice synthesis

After completing the aforementioned preprocessing steps, each MRI sample (a 3D image of size $256 \times 256 \times 256$) needs to be converted into a format suitable for input into the LNet model, which expects an input size of $224 \times 224 \times 3$. To address this, we extract 2D slices from the 3D volume, crop them, and stack three slices along the channel dimension to form a single input image.

To avoid duplicate usage of data and information leakage issues mentioned in related work, we adopt the following strategy:

Along the axial direction and separately for each axis, 45 continuous and equally spaced slices ($n = 1$) of size $256 \times 256 \times 1$ were sampled from the central region. For each sampling step, one coronal, one axial, and one sagittal slice were selected and stacked along the channel dimension to form a three-channel image with a size of $256 \times 256 \times 3$.

The dataset is split on the **sample level** (not slice level) to ensure that no slices from the same subject appear in both training and testing sets. The training and test sets are split at a ratio of 9:1, with class distributions of CN, MCI, and AD kept balanced (1:1:1) in both sets.

Further clarification regarding the “central region”: after spatial registration, the geometric centers of all brain images were aligned with the center of the adopted registration template. Therefore, within the unified coordinate space, the anatomical structures of all samples share consistent spatial correspondence. Based on this alignment, slices were selected within a fixed central sampling range to ensure that the brain slices used during training and validation are anatomically comparable and spatially consistent.

Each non-zero slice undergoes the following three preprocessing steps in sequence:

- Horizontal shifting to center the brain region in the image;
- Central cropping to a uniform size of $224 \times 224 \times 3$;
- Normalization to standardize the input distribution.

To improve model robustness and generalization, a 1:1 duplication of the training set is performed. The duplicated samples are processed using the following data augmentation techniques and mixed with the original training set for model training:

Table 1
Summary Statistics of Experimental Data.

Diagnosis	Number	Age	Gender: M/F	MMSE
CN	402	75.77 $\pm$ 5.14	211/191	29.21 $\pm$ 0.89
Dementia (AD)	390	75.16 $\pm$ 7.34	213/177	21.86 $\pm$ 4.35
MCI	401	74.59 $\pm$ 7.39	253/148	27.07 $\pm$ 2.02

Random cropping to $224 \times 224 \times 3$;
 Random rotation;
 Random horizontal and vertical flipping;
 Addition of Gaussian noise;
 Normalization.

Note: No data augmentation is applied to the test set—only standardized raw images are used to ensure fair evaluation.

As a result of the above pipeline, the training set contains a total of $1074 \times (45 + 45)$ multiview slices, while the test set contains approximately 119×45 slices (excluding all-zero slices, so actual numbers may vary slightly). All images are standardized to the input format of $224 \times 224 \times 3$ for model training and testing.

4.2. Experiments (experimental settings & evaluation metrics)

All experiments were conducted using the PyTorch framework on a server equipped with an NVIDIA RTX 3090 GPU. The study used four performance metrics: slice-level accuracy to assess the model's learning capability, image-level accuracy and Macro AUC to evaluate overall decision-making ability, and Balanced Accuracy to measure the model's recognition performance across the three classes. In addition, three efficiency and resource consumption metrics—GFLOPs, inference time per slice, and peak GPU memory usage—were introduced to assess the model's computational overhead. The image/slice-level accuracy is defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (20)$$

For class i ($i \in \{\text{AD}, \text{CN}, \text{MCI}\}$), the Balanced Accuracy (denoted as BA_i) is defined as:

$$BA_i = \frac{\text{Sensitivity}_i + \text{Specificity}_i}{2} = \frac{\frac{TP_i}{TP_i + FN_i} + \frac{TN_i}{TN_i + FP_i}}{2} \quad (21)$$

This metric statistically takes into account the model's ability to correctly identify both positive and negative samples for each class, rather than focusing solely on overall prediction accuracy, thereby providing a more comprehensive assessment of classification performance.

The detailed configuration of the experimental parameters is shown in Table 2.

To highlight the effectiveness of our method, we conducted comparative experiments with classic backbone networks, recently proposed high-performance models, and lightweight models. The table below shows the computational complexity of each model along with their performance on the current task. The computational complexity was calculated using the ptflops toolkit.

To further validate the actual contributions of the proposed improved modules across different versions, we conducted comparisons under the same dataset and training settings, evaluating the following configurations sequentially:

To evaluate the impact of input slices from different views on classification performance, we conducted experiments using the LSNet-t architecture with the following input configurations: single views

(axial (AX), coronal (COR), sagittal (SAG)), combined three views (AX + COR + SAG), and our proposed customized slicing scheme (Our Slices).

4.3. Discussion

From the comparative experimental results shown in Table 3, LSNet_t (the baseline model) demonstrated strong slice-level accuracy, reaching 75.11% (95% CI: 74.59, 75.63), ranking the third among all models. Its image-level accuracy was 84.20% (95% CI: 82.54, 85.86), showing good performance; combined with a Macro AUC of 96.68% (95% CI: 96.30, 97.06), the model exhibits strong overall decision-making capability. Although LSNet_t is slightly lower than some models in image-level decisions and recognition accuracy across the three classes. Its extremely low computational cost provides a clear advantage in resource efficiency: GFLOPs are only 0.64, and peak GPU memory usage is 1063.01 MB, significantly lower than the lightweight StarNet_S1 (GFLOPs: 0.86, peak memory: 2545.10 MB) and EfficientNet_B0 (GFLOPs: 0.81, peak memory: 5494.23 MB). Compared to these two models, LSNet_t achieves only 74.42% and 79.01% of their GFLOPs, and 41.77% and 19.35% of their peak memory usage, while its slice-level accuracy is 2.57% and 2.68% higher, with Macro AUC also more favorable. By contrast, although EfficientNet_B3 achieves some improvement in overall performance, its slice-level accuracy and Macro AUC remain lower than LSNet_t, and this performance gain comes at the cost of $3.13 \times$ GFLOPs and $10.06 \times$ peak memory, resulting in a lower cost-effectiveness. StarNet_S3 performs worse than S1 across all metrics and is therefore not discussed further. It is worth noting that the slice-level inference time of LSNet_t is 2.16 ms, which is at a moderate level, and the image-level inference time shows a similar trend. Unlike GFLOPs and peak memory, which are highly hardware-dependent, inference time is a relatively "soft" metric, minimally affected by hardware, requiring only a short waiting time for computation. Given the unit is milliseconds, unless differences are several times larger (For example, in ASPP, the slice-level and image-level inference times are 6.35 ms and 440.82 ms, respectively), the impact on overall inference overhead is limited.

- Left column: class-level ROC curves, with the x-axis as False Positive Rate and the y-axis as True Positive Rate;
- Middle column: class-level Precision-Recall curves, with the x-axis as Recall and the y-axis as Precision;
- Right column: class-level calibration curves, with the x-axis as Predicted Probability and the y-axis as Fraction of positives.

(a) LSNet_t; (b) Inception_v3; (c) LSNet_t + FWFA.

Considering performance, LSNet_t exemplifies the "lightweight and efficient" characteristics defined in this study and was selected as the backbone network, with further optimization performed to address observed shortcomings. In comparison, Inception_v3 ranks first in slice-level accuracy, image-level accuracy, class-wise Balanced Accuracy, and Macro AUC. Although its high computational cost limits its advantage in hardware-constrained environments, its superior performance serves as a stronger medical baseline, providing a robust benchmark to validate the effectiveness of our optimizations in comparative experiments.

We designed three optimization schemes for LSNet and compared them with related works in module ablation experiments. The ablation experiments were divided into four parts: baseline, channel attention, spatial attention, and multi-attention fusion, with the baseline using the original LSNet. Table 4 reports the results of the ablation experiments.

At the channel attention level, we compared three designs: ECA_var2, ECA_var3, and MECA. Overall, all three channel attention modules improved model performance comprehensively, while GFLOPs remained nearly unchanged and peak memory usage increased by less than 40 MB. MECA achieved slightly superior overall performance: slice-level accuracy of 76.54% (95% CI: 75.41, 77.67), image-level accuracy of 88.57% (95% CI: 86.05, 91.09), CN Balanced Accuracy of 96.10%

Table 2
Experimental Parameters.

Optimizer	Adamw
batch size	256
training epochs	150
LR schedule	cosine
max learning rate	$2e-3$
min learning rate	$5e-5$
weight decay	0.25
early stopping	20

Table 3
Comparative Experiments Among Different Networks.

Model	GFLOPs	Inference time per slice (ms)	Inference time per image (ms)	Peak GPU memory usage (MB)	Slice-level Accuracy (%)		Image-level Accuracy (%)		Balanced Accuracy(%)						Macro AUC	
					Mean	95% CI	Mean	95% CI	CN		AD		MCI		Mean	95% CI
									Mean ± Std	95% CI	Mean ± Std	95% CI	Mean ± Std	95% CI		
LSNet_t(baseline)	0.64	2.16	150.15	1063.01	75.11	(74.59, 75.63)	84.20	(82.54, 85.86)	93.34	(91.74, 94.94)	87.65	(85.25, 90.05)	83.43	(81.20, 85.66)	96.68	(96.30, 97.06)
ResNet18 [34]	3.65	1.41	98.49	1598.40	73.70	(72.53, 74.87)	84.54	(81.85, 87.23)	94.72	(92.85, 96.59)	88.20	(85.64, 90.76)	82.33	(85.34, 91.12)	96.46	(95.94, 96.98)
SE-ResNet18 [35]	3.65	1.55	108.23	1778.11	74.08	(72.95, 75.21)	85.55	(82.50, 88.60)	95.22	(94.15, 96.29)	89.75	(88.14, 91.36)	82.59	(77.82, 87.36)	96.53	(95.80, 97.26)
Inception_V3 [36]	5.71	2.72	183.05	3499.79	78.88	(77.09, 80.67)	89.92	(86.83, 93.01)	96.09	(93.81, 98.37)	92.37	(89.96, 94.78)	88.84	(85.71, 91.97)	98.51	(98.11, 98.91)
StarNet_S1 [37]	0.86	1.79	124.57	2545.10	72.54	(71.47, 73.61)	84.84	(83.91, 85.77)	95.60	(94.49, 96.71)	87.57	(86.48, 88.66)	82.82	(81.22, 84.42)	95.81	(95.47, 96.15)
StarNet_S3 [37]	1.53	2.06	143.24	3367.84	72.40	(71.73, 73.07)	81.01	(79.59, 82.43)	92.33	(90.23, 94.43)	84.36	(81.21, 87.51)	80.56	(79.28, 81.84)	95.47	(95.26, 95.68)
EfficientNet_B0 [38]	0.81	2.23	156.84	5494.23	72.43	(71.40, 73.46)	82.02	(80.77, 83.27)	92.20	(90.76, 93.64)	85.62	(82.21, 89.03)	81.68	(79.41, 83.95)	96.25	(95.90, 96.60)
EfficientNet_B3 [38]	2.00	3.74	260.14	10698.50	73.15	(72.54, 73.76)	84.37	(82.32, 86.42)	93.71	(92.01, 95.41)	86.12	(84.85, 87.39)	84.94	(81.71, 88.17)	96.31	(95.96, 96.66)
Densenet121 [39]	5.74	3.15	219.18	8119.11	77.04	(75.93, 78.15)	87.73	(85.48, 89.98)	95.86	(94.33, 97.39)	89.71	(87.16, 92.26)	86.82	(83.53, 90.11)	97.54	(97.07, 98.01)
Visformer_tiny [40]	2.42	1.85	128.95	2668.88	69.07	(67.02, 71.12)	81.85	(80.02, 83.68)	92.45	(90.39, 94.51)	85.14	(83.00, 87.28)	81.57	(79.30, 83.84)	95.24	(94.78, 95.70)
LSNet t + MECA	0.64	2.25	156.25	1090.92	76.54	(75.41, 77.67)	88.57	(86.05, 91.09)	96.10	(94.51, 97.69)	89.18	(86.77, 91.59)	88.95	(85.10, 92.80)	97.55	(97.17, 97.93)
LSNet t + MASA	0.82	2.91	195.86	1820.75	76.39	(75.42, 77.36)	86.72	(83.45, 89.99)	94.97	(92.31, 97.63)	88.68	(86.85, 90.51)	86.45	(82.81, 90.10)	97.61	(97.01, 98.21)
LSNet t + FWFA	0.82	3.31	230.04	1458.53	78.14	(76.86, 79.42)	90.08	(88.86, 91.30)	96.85	(95.99, 97.71)	91.87	(90.50, 93.24)	88.96	(87.52, 90.40)	98.33	(98.06, 98.60)

Table 4
Ablation Study.

Model	GFLOPs	Inference time per slice (ms)	Inference time per image (ms)	Peak GPU memory usage (MB)	Slice-level Accuracy (%)		Image-level Accuracy (%)		Balanced Accuracy(%)			Macro AUC		
					Mean	95% CI	Mean ± Std	95% CI	CN	Mean	95% CI	AD	Mean	95% CI
LSNet t (baseline)	0.64	2.16	150.15	1063.01	75.11	(74.59, 75.63)	84.20	(82.54, 85.86)	93.34	91.74,	(85.25, 90.05)	87.65	83.43	(81.20, 85.66)
LSNet t + ECA_var2	0.64	2.07	144.48	1056.02	76.14	(75.52, 76.76)	87.39	(85.20, 89.58)	95.10	92.98,	(88.68, 90.96)	89.82	86.69	(83.60, 89.78)
LSNet t + ECA_var3	0.64	2.16	145.37	1073.59	76.09	(74.91, 77.27)	88.07	(85.08, 91.06)	95.36	93.56,	(86.96, 93.18)	90.07	87.69	(84.70, 90.68)
LSNet t + MECA	0.64	2.25	156.25	1090.92	76.54	(75.41, 77.67)	88.57	(86.05, 91.09)	96.10	94.51,	(86.77, 91.59)	89.18	88.95	(85.10, 92.80)
LSNet t + Pyramid	1.67	2.91	223.92	1345.08	75.76	(74.65, 76.87)	87.73	(86.66, 88.80)	94.85	92.91,	(87.81, 92.13)	89.97	86.08	(84.43, 87.73)
LSNet t + ASPP(32)	2.42	6.35	440.82	1898.78	73.42	(72.04, 74.80)	83.36	(82.14, 84.58)	93.33	91.65,	(86.02, 87.56)	86.79	82.44	(80.95, 83.93)
LSNet t + MASA	0.82	2.91	195.86	1820.75	76.39	(75.42, 77.36)	86.72	(83.45, 89.99)	94.97	92.31,	(86.85, 90.51)	88.68	86.45	(82.81,90.10)
LSNet t + CBAM(33)	0.64	2.39	165.66	1090.38	73.17	(72.54, 73.80)	83.70	(82.14, 85.26)	93.83	92.58,	(84.70, 89.36)	87.03	82.44	(79.34, 85.54)
LSNet t + FWFA	0.82	3.31	230.04	1458.53	78.14	(76.86, 79.42)	90.08	(88.86, 91.30)	96.85	95.99,	(90.50, 93.24)	91.87	88.96	(87.52, 90.40)

(95% CI: 94.51, 97.69), and MCI Balanced Accuracy of 88.95% (95% CI: 85.10, 92.80), achieving the best results across multiple metrics. In general, channel attention modifications outperformed the baseline, validating that the dimensionality reduction operation in the original LSNet channel attention negatively affects learning and decision-making, and that the proposed optimization is effective. These results also confirm that introducing multi-granular fusion in ECA can further improve overall model performance.

For spatial attention, we compared Pyramid, ASPP, and MASA designs. Experimental results show that traditional multi-scale spatial modeling methods (Pyramid, ASPP) slightly enhance spatial perception, but with significant increases in GFLOPs and inference time, their performance gains are limited, and ASPP even shows degradation in slice-level and image-level accuracy. In contrast, MASA demonstrates the best cost-effectiveness: slice-level accuracy of 76.39% (95% CI: 75.42, 77.36), 0.63% higher than Pyramid; image-level decision accuracy slightly lower than Pyramid by 1.01%, but GFLOPs are only half. By replacing block pooling + dilated convolution and convolutional feature fusion with standard + dilated convolution multi-scale receptive fields, combined with a spatial selection mechanism, MASA significantly reduces computational overhead while maintaining good performance.

At the fusion attention level, we compared the traditional CBAM (sequential structure) with FWFA (parallel weighted fusion structure). The results show that FWFA significantly outperforms CBAM and the baseline across all key metrics, confirming that carefully designed multi-attention weighted fusion can achieve better performance than simple sequential dual attention.

Under the constraint of a lightweight design, FWFA achieves comprehensive performance improvement at the cost of increasing baseline GFLOPs by 28.13%, peak memory usage by 27.12%, and slice inference time by 53.24%. These experimental results fully demonstrate that multi-granularity, multi-receptive field, and multi-attention fusion mechanisms can more comprehensively capture lesion features in MRI brain images, enhancing the model's learning and decision-making capabilities. Specifically, FWFA significantly outperforms the baseline and other attention module ablations in slice-level accuracy, image-level accuracy, Macro AUC, and class-wise Balanced Accuracy. This indicates that by fusing multi-scale features with channel and spatial attention information, the model can more effectively focus on key lesion regions, suppress redundant information, and strengthen the representation of complex pathological signals, achieving more intelligent and reliable recognition and decision-making.

Compared with Inception_v3, the FWFA-enhanced LSNet shows slightly lower slice-level accuracy (−0.74%), Macro AUC (−0.22), and AD Balanced Accuracy (−0.5%), but slightly higher image-level accuracy (+0.16%) and improved CN and MCI Balanced Accuracy (+0.79% and +0.12%, respectively). This indicates that deeper and more computationally expensive networks do achieve better learning and feature recognition, yet Inception_v3's GFLOPs and peak memory usage are 696.34% and 239.95% of LSNet + FWFA, reflecting its high-cost, high-performance nature. We further visualized selected evaluation metrics for the baseline, Inception_v3, and LSNet t + FWFA models. Fig. 6 presents the confusion matrices of the three networks, while Fig. 7 displays their ROC curves, PR curves, and calibration plots side by side for direct comparison.

Additionally, we further compare the proposed synthetic slice pipeline in this study with the conventional single-view slicing methods, as shown in Table 5. These experiments were all conducted using the original LSNet_t network, therefore, GFLOPs, slice-level inference time, and peak GPU memory usage showed no significant differences. The method of slice composition has a slight effect on image-level inference time, with the conventional approach showing slightly lower overhead than our method. However, since the unit is in milliseconds, this difference has negligible impact in practical applications. The results show that our method significantly outperforms all single-view slice approaches in overall performance. Although recognition for CN and MCI

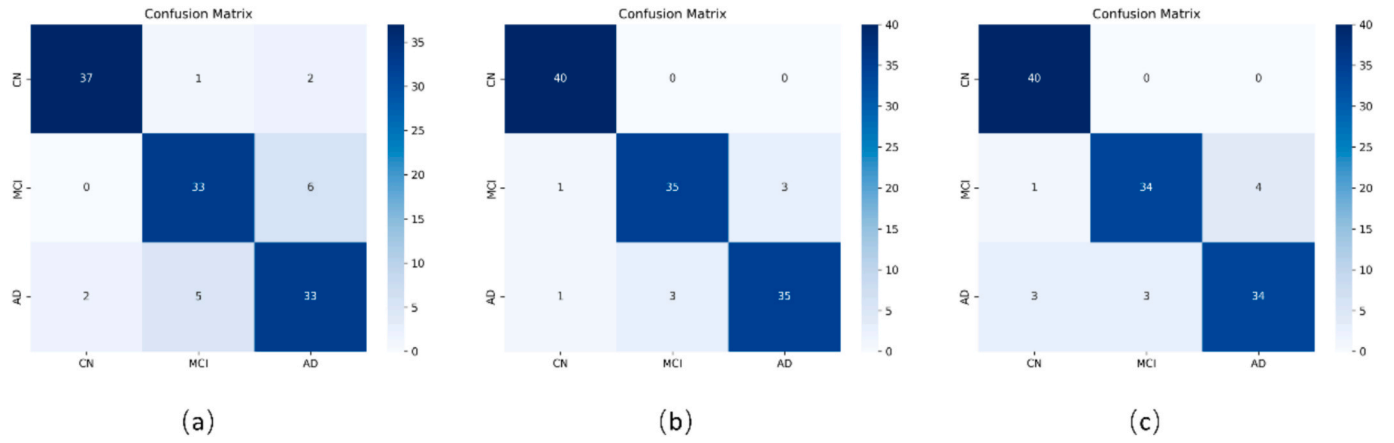


Fig. 6. Comparison of classification results of different networks on the test set; (a) LSNet_t; (b) Inception_v3; (c) LSNet_t + FWFA.

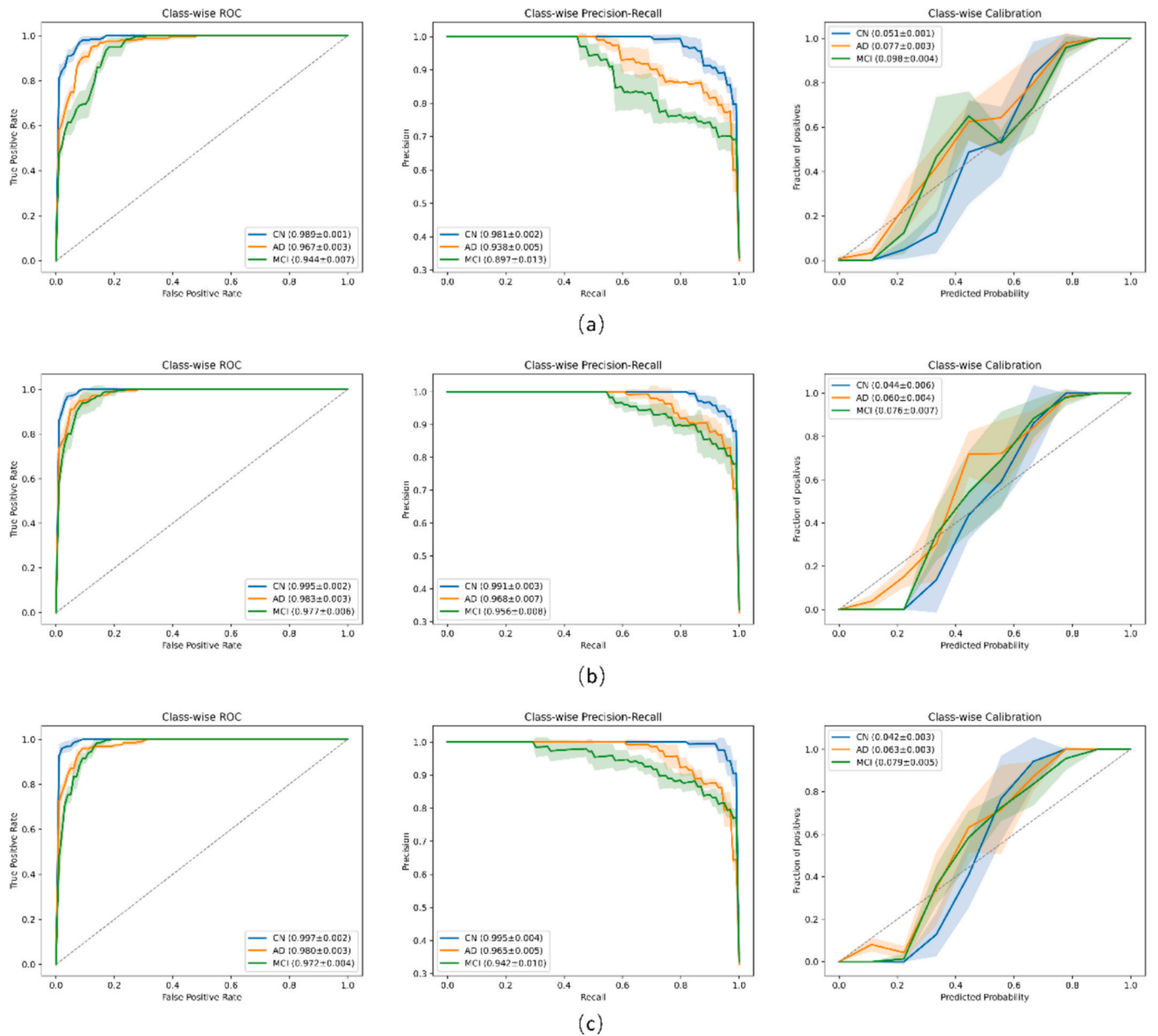


Fig. 7. Visualization of classification performance of different networks on the test set. Each row shows three types of curves for the same network, with curve colors representing different classes (CN, AD, MCI):

Table 5
Comparative Experiments of Different Data Utilization Methods.

Slicing Strategy	GFLOPs	Inference time per slice (ms)	Inference time per image (ms)	Peak GPU memory usage (MB)	Slice-level Accuracy (%)		Image-level Accuracy (%)		Balanced Accuracy(%)						Macro AUC	
					Mean	95% CI	Mean	95% CI	CN	AD		MCI	Mean	95% CI	Mean	95% CI
										Mean	95% CI					
										Mean	95% CI					
Our Strategy	0.64	2.16	150.15	1063.01	75.11	(74.59, 75.63)	84.20	(82.54, 85.86)	93.34	87.65	87.65 ± 1.93	83.43	(81.20, 85.66)	96.68	(96.30, 97.06)	
Only AX	0.64	2.13	143.98	1062.55	65.36	(63.61, 67.11)	79.66	(77.66, 81.66)	92.9	83.96	83.96 ± 1.13	82.85	(79.45, 86.25)	93.41	(92.79, 94.03)	
Only COR	0.64	2.18	142.46	1064.69	60.32	(59.72, 60.92)	78.49	(77.07, 79.91)	91.97	82.55	82.55 ± 1.27	83.24	(80.37, 86.11)	91.97	(90.79, 93.15)	
Only SAG	0.64	2.18	149.89	1066.40	62.61	(61.82, 63.40)	79.16	(77.94, 80.38)	91.69	84.96	84.96 ± 0.84	82.22	(80.66, 83.78)	92.92	(92.22, 93.62)	
AX + COR + SAG	0.64	2.15	144.91	1067.49	68.56	(67.53, 69.59)	81.85	(81.02, 82.68)	94.09	86.41	86.41 ± 0.67	84.11	(82.36, 85.86)	94.75	(94.32, 95.18)	

is slightly lower by 0.75% and 1.43%, slice-level and image-level accuracy are higher by 6.55% and 2.35%, demonstrating clear advantages.

This study performs training and validation based on two-dimensional image slices, where slice-level accuracy directly reflects the model's learning capability. Meanwhile, we acknowledge that clinical decisions should rely on the entire image rather than individual slices; therefore, an image-level decision strategy was designed. Specifically, image-level predictions are obtained by averaging the softmax probabilities of all slices. This aggregation is essentially a posterior statistical integration process, introducing no additional learnable parameters or new feature modeling. Thus, improvements at the image level stem from the statistical stabilization effect of probability averaging, which reduces the influence of slice-level prediction fluctuations on the final decision.

Although this approach captures the overall trend of slice predictions, it does not fully align with the network's intrinsic behavior. The adopted image-level aggregation is not a strict multiple instance learning framework, but a practical approximation built upon the original slice-based training paradigm. Nevertheless, this does not undermine the reliability of image-level metrics—slice-level and image-level evaluations reflect model performance from theoretical and clinical perspectives, respectively, and are complementary.

There remains room for further improvement. Learning global pathological distribution patterns beyond slice-level decisions to build a more interpretable image-level framework is a meaningful direction for future research. We plan to explore explicit subject-level modeling strategies, such as multiple instance learning or learnable slice-weighting mechanisms, to further enhance subject-level prediction performance while preserving slice-level discriminative ability.

All experimental results show a consistent trend in class-wise Balanced Accuracy: CN is the easiest to recognize, followed by AD, with MCI being the most challenging. This aligns with MCI as an intermediate state between CN and AD, exhibiting pathological features of both groups. From an anatomical perspective, AD-related neurodegeneration mainly affects the hippocampus and medial temporal lobe structures (e.g., parahippocampal gyrus and temporal pole), as well as the default mode network (DMN, including prefrontal cortex, posterior cingulate cortex, precuneus, etc.), which show marked volume loss and decreased functional connectivity in AD patients.

Ledig et al. [41] reported that as Alzheimer's disease progresses, medial temporal structures (e.g., hippocampus and entorhinal cortex) exhibit significant atrophy; cross-sectional comparisons further show that hippocampal volume loss in MCI subjects lies between cognitively normal controls and AD patients. RomBouts et al. [42] found in functional imaging studies that DMN deactivation during visual encoding or working memory tasks progresses from healthy controls (HC) to MCI to AD, with MCI exhibiting intermediate deactivation, supporting the view of MCI as a transitional state from normal cognition to AD. Therefore, MCI displays some AD pathological features while retaining aspects of healthy status, which likely contributes to its greatest difficulty in model classification tasks. Future work will further explore strategies to improve MCI recognition.

5. Conclusion

This study significantly improved the diagnostic performance of the baseline network while maintaining low computational overhead and preserving model lightweight characteristics. The model's low resource dependency facilitates deployment in various healthcare settings, providing strong technical support for clinical research and auxiliary diagnosis of Alzheimer's disease. Additionally, a novel 2D MRI slice synthesis method was proposed. Experimental results show that compared to traditional single-view slice methods, this approach carries richer structural information, greatly enhancing feature learning and overall performance, demonstrating promising application potential.

CRediT authorship contribution statement

Jinhua Sheng: Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Ruilin Huang:** Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Qiao Zhang:** Validation, Project administration, Methodology. **Tao Li:** Data curation. **Miao Miao:** Formal analysis. **Yan Lu:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (CN) under Grant No.62271177 and the Key Program of Natural Science Foundation of Zhejiang Province (CN) under Grant No. LZ24F010007.

Ethical statement

This study was approved by the institutional review board (IRB) at Hangzhou Dianzi University (IRB-2020001) and the ethics committee at Beijing Hospital (2022BJYYEC-375-01).

Data availability

The raw data and participant scans can be found in from the Alzheimer's Disease Neuroimaging Initiative (ADNI) (<https://adni.loni.usc.edu/>). Additional information can be provided by the corresponding author upon reasonable request.

References

- [1] World Health Organization. Dementia [EB/OL]. 2023-03-15 [2023-07-10].
- [2] National Institute on Aging. Alzheimer's Disease Fact Sheet [EB/OL]. Archived 2022-03-23 [Retrieved 2022-03-23].
- [3] A. Association, 2024 Alzheimer's disease facts and figures, *Alzheimer's & Dementia* 20 (5) (2024) 3708–3821, <https://doi.org/10.1002/alz.13809>.
- [4] F. Bunn, A.M. Burn, C. Goodman, et al., Systematic review of information and support interventions for caregivers of people with dementia, *BMC Geriatr.* 7 (2007) 18, <https://doi.org/10.1186/1471-2318-7-18>.
- [5] S. Khan, K.H. Barve, M.S. Kumar, Recent advancements in pathogenesis, diagnostics and treatment of Alzheimer's disease, *Curr. Neuropharmacol.* 18 (11) (2020) 1106–1125.
- [6] K. Gauthreaux, T.A. Bonnett, L.M. Besser, et al., Concordance of clinical Alzheimer diagnosis and neuropathological features at autopsy, *J. Neuropathol. Exp. Neurol.* 79 (2020).
- [7] National Institute for Health and Clinical Excellence. Dementia: Quick Reference Guide. London: NICE, 2006.
- [8] R. Zhang, J. Sheng, Q. Zhang, et al., A review of multimodal fusion-based deep learning for Alzheimer's disease, *Neuroscience* 576 (2025) 80–95, <https://doi.org/10.1016/j.neuroscience.2025.04.035>.
- [9] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, *Nature* 521 (7553) (2015) 436–444.
- [10] Y. Chen, T. Luo, S. Liu, J. Hu, Deep learning on computational-resource-limited platforms: a survey, *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.* 39 (11) (2020) 3325–3338, <https://doi.org/10.1109/TCAD.2019.2949514>.
- [11] R. Xiao, et al., Early diagnosis model of Alzheimer's disease based on sparse logistic regression with the generalized elastic net, *Biomed. Signal Process. Control* 66 (2021) 102362, <https://doi.org/10.1016/j.bspc.2020.102362>.
- [12] A. Khan, S. Zubair, An improved multi-modal based machine learning approach for the prognosis of Alzheimer's disease, *J. King Saud Univ. – Comput. Info. Sci.* 34 (6) (2022) 2688–2706, <https://doi.org/10.1016/j.jksuci.2020.04.004>.
- [13] S. Basheera, S.S. Ram, M., Leung-Malik features and Adaboost perform classification of Alzheimer's disease stages, *IETE J. Res.* (2022), <https://doi.org/10.1080/03772063.2022.2154284>.
- [14] Q. Zhu, et al., Classification of Alzheimer's disease based on abnormal hippocampal functional connectivity and machine learning, *Front. Aging Neurosci.* 14 (2022) 754334, <https://doi.org/10.3389/fnagi.2022.754334>.
- [15] H. Lao, X. Zhang, Regression and classification of Alzheimer's disease diagnosis using NMF-TDNet features from 3D brain MR image, *IEEE J. Biomed. Health Inform.* 26 (3) (2022) 1103–1115, <https://doi.org/10.1109/JBHI.2021.3113668>.
- [16] Özçelik Y B, Altan A, Kaya C. Machine learning approach for early diagnosis of alzheimer's disease using rs-fMRI and metaheuristic optimization with functional connectivity matrices. In: *2024 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET)*, 2024: 248–253. doi:10.1109/IICAIET62352.2024.10730420.
- [17] H. Sun, A. Wang, W. Wang, C. Liu, An improved deep residual network prediction model for the early diagnosis of Alzheimer's disease, *Sensors* 21 (12) (2021) 4182, <https://doi.org/10.3390/s21124182>.
- [18] S. Fathi, A. Ahmadi, A. Dehnad, et al., A deep learning-based ensemble method for early diagnosis of Alzheimer's disease using MRI images, *Neuroinformatics* 22 (2024) 89–105, <https://doi.org/10.1007/s12021-023-09646-2>.
- [19] A. Ebrahimi-Ghahnavieh, S. Luo, R. Chiong, Transfer Learning for Alzheimer's Disease Detection on MRI Images, in: *Proceedings of the IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communications Technology (IAICT)*, 2019, pp. 133–138, <https://doi.org/10.1109/IICTAICT.2019.8784845>.
- [20] N. Zeng, H. Li, Y. Peng, A new deep belief network-based multi-task learning for diagnosis of Alzheimer's disease, *Neural Comput. Applic.* (2021), <https://doi.org/10.1007/s00521-021-06149-6>.
- [21] Z. Fan, et al., U-net based analysis of MRI for Alzheimer's disease diagnosis, *Neural Comput. Applic.* 33 (20) (2021) 13587–13599, <https://doi.org/10.1007/s00521-021-05983-y>.
- [22] S. Katabathula, Q. Wang, R. Xu, Predict Alzheimer's disease using hippocampus MRI data: a lightweight 3D deep convolutional network model with visual and global shape representations, *Alzheimer's Res. Therapy* 13 (1) (2021) 9, <https://doi.org/10.1186/s13195-021-00837-0>.
- [23] W. Yu, B. Lei, M.K. Ng, et al., Tensorizing GAN with high-order pooling for Alzheimer's disease assessment, *IEEE Trans. Neural Netw. Learn. Syst.* 33 (9) (2022) 4945–4959, <https://doi.org/10.1109/TNNLS.2021.3063516>.
- [24] H. Guan, C. Wang, J. Cheng, J. Jing, T. Liu, A parallel attention-augmented bilinear network for early magnetic resonance imaging-based diagnosis of Alzheimer's disease, *Hum. Brain Mapp.* 43 (2) (2022) 760–772, <https://doi.org/10.1002/hbm.25685>.
- [25] Kina E. TLEABLCNN: Brain and Alzheimer's Disease Detection Using Attention-Based Explainable Deep Learning and SMOTE Using Imbalanced Brain MRI [J]. In: *IEEE Access*, 2025, 13: 27670–27683. doi: 10.1109/ACCESS.2025.3539550.
- [26] M. Abdelaziz, T. Wang, W. Anwaar, A. Elazab, Multi-scale multimodal deep learning framework for Alzheimer's disease diagnosis, *Comput. Biol. Med.* 184 (2025) 109438, <https://doi.org/10.1016/j.cmpbiomed.2024.109438>.
- [27] Hu Z, Wang Z, Jin Y, Hou W. VGG-TSwinformer: Transformer-based deep learning model for early Alzheimer's disease prediction. In: *Computer Methods and Programs in Biomedicine*, 2023, 229: 107291. doi: 10.1016/j.cmpb.2022.107291.
- [28] Wang A, Chen H, Lin Z, Han J, Ding G. LSNNet: See Large, Focus Small. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025: 9718–9729.
- [29] Q. Wang, B. Wu, P. Zhu, P. Li, W. Zuo, Q. Hu, ECA-Net: Efficient Channel Attention for Deep Convolutional Neural Networks. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [30] A. Kassner, R.E. Thornhill, Texture analysis: a review of neurologic MR imaging applications, *Am. J. Neuroradiol.* 31 (5) (2010) 809–816, <https://doi.org/10.3174/ajnr.A2061>.
- [31] H. Zhao, J. Shi, X. Qi, et al., Pyramid Scene Parsing Network [C], in: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2881–2890, <https://doi.org/10.1109/CVPR.2017.660>.
- [32] L.C. Chen, G. Papandreou, I. Kokkinos, et al., Deeplab: semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs, *IEEE Trans. Pattern Anal. Mach. Intell.* 40 (4) (2018) 834–848, <https://doi.org/10.1109/TPAMI.2017.2699184>.
- [33] S. Woo, J. Park, J.Y. Lee, et al., CBAM: Convolutional Block Attention Module, in: *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 3–19, https://doi.org/10.1007/978-3-030-01234-2_1.
- [34] K. He, X. Zhang, S. Ren, J. Sun, Deep Residual Learning for image Recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [35] J. Hu, L. Shen, G. Sun, Squeeze-and-Excitation Networks. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- [36] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, Z. Wojna, Rethinking the Inception Architecture for Computer Vision. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [37] Ma X, Dai X, Bai Y, Wang Y, Fu Y. Rewrite the Stars. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024: 5694–5703.
- [38] Tan M, Le Q. EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks. In: *Proceedings of the 36th International Conference on Machine Learning (ICML)*, 2019: 6105–6114.
- [39] G. Huang, Z. Liu, L. Van Der Maaten, et al., Densely Connected Convolutional Networks, in: *Proceedings of the IEEE Conference on Computer Vision and Pattern*

- Recognition (CVPR)*, 2017, pp. 4700–4708, <https://doi.org/10.1109/CVPR.2017.243>.
- [40] Chen Z, Xie L, Niu J, Liu X, Wei L, Tian Q. Visformer: The Vision-Friendly Transformer. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021: 589–598.
- [41] C. Ledig, A. Schuh, R. Guerrero, et al., Structural brain imaging in Alzheimer's disease and mild cognitive impairment: biomarker analysis and shared morphometry database, *Sci. Rep.* 8 (2018) 11258, <https://doi.org/10.1038/s41598-018-29295-9>.
- [42] S.A. Rombouts, F. Barkhof, R. Goekoop, C.J. Stam, P. Scheltens, Altered resting state networks in mild cognitive impairment and mild Alzheimer's disease: an fMRI study, *Hum. Brain Mapp.* 26 (4) (2005) 231–239, <https://doi.org/10.1002/hbm.20160>.